

AI TRAINING – WEEK 2

ASSIGNMENT 1

Lists, Tuples & Sets

Learn → Practice → Build → Explain

1. What You Will Learn

You will learn three important Python collection types: lists, tuples, and sets. You will practice accessing and modifying them, identify common errors, and build two small real-world utilities.

- Lists – indexing, slicing, methods, and mutability
- Tuples – indexing, slicing, methods, and immutability
- Sets – unique values and common set operations
- Difference between list, tuple, and set
- Common errors and how to fix them
- Build a simple shopping cart
- Remove duplicate values using a set

2. Student Deliverable – Keep It Simple

Submit ONE Jupyter Notebook only.

File name: Week2_Day1_Lists_Tuples_Sets_YourName.ipynb

Your notebook should contain:

- Part A – Basic practice
- Part B – Lists
- Part C – Tuples
- Part D – Sets
- Part E – Debugging
- Part F – Shopping Cart mini project
- Part G – Duplicate Removal utility
- Part H – Explain what you learned

No separate Python files, Word/PDF files, or screenshots are required.

3. Part A – Understand the Basics

Create and print:

```
fruits = ["apple", "banana", "orange"]
coordinates = (10, 20)
numbers = {1, 2, 3, 4}
```

- What is a list?
List is collection elements in orderly.it contain different type of datatypes .It is mutable
- What is a tuple?

Tuple is also a collection of elements in orderly and contain different datatypes.it is immutable

- What is a set?

Set is container.it can't allow duplicates.it was less in bulid function then compare to list and tuple

- Which one allows duplicate values?
- Which one is mutable?

4. Part B – List Practice

```
fruits = ["apple", "banana", "orange", "mango", "grapes"]
```

- Print the first item.
`print(fruits[0])----->apple`
- Print the last item.
`print(fruits[-1])----->grapes`
- Print the first three items using slicing.
`Print(fruits[:4])----->["apple","banana","orange"]`
- Change "banana" to "watermelon".
`Fruits[1]="watermelon"`
- Add "kiwi".
`Fruits.append("kiwi")`
- Remove "orange".
`Fruits.remove("orange")`
- Print the final list.
`["apple","watermelon","mango","grapes"]`
- Find the length of the list.
`Len(fruits)----->4`

Practice: `append()`, `insert()`, `remove()`, `pop()`, `sort()`, `reverse()`, `len()`

5. Part C – Tuple Practice

```
student = ("Phani", 34, "Hyderabad", "Python")
```

- Print the first item.
`Print(student[0])----->"phani"`
- Print the last item.
`Print(student[-1])----->"python"`
- Print the first two items using slicing.
`Print(student[:3])----->("phani",34)`
- Find the length.
`Print(len(student))----->4`
- Try changing 34 to 35.
It came error
- Observe the error and explain why it happened.
Because tuple is immutable

Remember: tuples are generally used when the collection should not be changed.

6. Part D – Set Practice

```
numbers = {10, 20, 30, 20, 10, 40}
```

- Print the set.
Print(numbers)----->{10,20,30}
- Observe what happened to duplicates.
Duplicates are removed.the set can't allow the duplicates
- Add 50.
Numbers.add(50)----->{10,20,30,50}
- Remove 20.
Numbers.remove(20)----->{10,30,50}
- Check whether 30 exists.
30 in Numbers
- Find the number of unique values.
Len(numbers)
Set have already unique values.soooo we just count them

Important: sets are unordered collections. Do not depend on a particular display order.

7. Part E – List vs Tuple vs Set

Feature	List	Tuple	Set
Ordered?	yes	yes	no
Allows duplicates?	yes	yes	no
Mutable?	yes	no	yes
Supports indexing?	yes	yes	no
Main use?	For duplicates and changeable values	Unchangeable values	Unique values

8. Part F – Quick Debugging

Find the error, fix it, and explain the mistake.

Debugging Problem 1

```
fruits = ["apple", "banana", "orange"]
print(fruits[5])----->error not of boundary

Print(fruits[2])
```

Debugging Problem 2

```
numbers = (10, 20, 30)
numbers[0] = 100----->error can't change the tuple
```

Debugging Problem 3

```
items = {1, 2, 3}
print(items[0])----->error the set not allow the indexing
```

Debugging Problem 4

```
fruits = ["apple", "banana"]
fruits.add("orange")----->add function for set for list it is append
```

9. Part G – Hands-on Activity 1: Shopping Cart

Start with: `cart = ["Rice", "Milk", "Bread"]`

Build a simple shopping cart using a list. Your program should:

- Display the cart.

`Print(cart)----->["Rice", "Milk", "Bread"]`

- Add a new item.

Error because it is tuple

- Remove an item.

Error because it is a tuple

- Display the number of items.

`Len(cart)`

- Check whether a particular item exists.

`"milk" in cart`

- Display the final cart.

`Print(cart)----->["Rice", "Milk", "Bread"]`

Bonus: create `add_item()`, `remove_item()`, and `show_cart()` functions.

10. Part H – Hands-on Activity 2: Duplicate Removal

Create a program that removes duplicate values from a list using a set.

Example input: `[10, 20, 10, 30, 20, 40, 30]`

Expected result: a collection containing only unique values.

Test with:

- `[1, 1, 2, 2, 3, 3]`
- `["apple", "banana", "apple", "orange", "banana"]`
- `[10, 20, 30]`

`List1=[10, 20, 10, 30, 20, 40, 30]`

`Set1=set(list1)`

`Print(Set1)`

Output:-

`{40,10,20,30}`

Think: Why is a set useful for this problem?

To remove the duplicates.

11. Part I – Explain in Your Own Words

1. What is the main difference between a list and a tuple?

Answer: list is mutable and tuple is immutable

List with []

Tuple with ()

2. Why can't we normally change an item inside a tuple?

Answer: because tuple is immutable

3. Why are sets useful for removing duplicates?

Answer: it can't allow the duplicate

4. What does indexing mean?

Answer: assigning the position for each element

5. What does slicing mean?

Answer: sub part from the container.

6. What does mutable mean?

Answer: it allows the changes.

7. When would you choose a list instead of a tuple?

Answer: when the changes are required in the future

12. Student Submission Checklist

- I completed all practice sections.✓
- I tested my code.✓
- I fixed the debugging problems.✓
- I completed the Shopping Cart activity.✓
- I completed the Duplicate Removal activity.✓
- I explained the concepts in my own words.✓
- My notebook runs without major errors.✓
- My filename is correct.✓

13. After This Assignment, You Should Be Able To

- Create and use lists, tuples, and sets.
- Access values using indexes.
- Use slicing.
- Use common collection methods.
- Understand mutability and immutability.
- Remove duplicates using a set.
- Choose the right collection for a simple problem.

Key Principle: Don't just memorize syntax. Understand when and why to use each collection.